TryHackMe - Nmap Basic Port Scans

Platform: TryHackMe
Room link: Nmap Basic Port Scans

1. Introduction

Port scanning is an essential part of network reconnaissance and penetration testing. It helps identify which ports are open, closed, or filtered on a target system.

Nmap provides several scanning techniques that allow security professionals to discover exposed network services and understand how a target responds to different network requests.

In this room, we explore TCP and UDP port scanning, examine different port states, and learn how to perform service enumeration using Nmap.

2. Objectives

The main objectives of this room are:

- Understand the purpose of port scanning.
- Learn how TCP connections work.
- Perform TCP Connect scans.
- Understand TCP SYN scans.
- Perform UDP scans.
- Identify different port states.
- Learn how to scan specific ports and port ranges.
- Understand service and version detection.
- Analyse scan results.
- Save and review Nmap scan output.

3. Tools and Requirements

Tool	Purpose
Nmap	Network and port scanning
Kali Linux	Penetration testing environment
TryHackMe AttackBox	Browser-based Linux environment
Wireshark	Network traffic analysis
Linux Terminal	Executing commands

4. Task-wise Walkthrough

Task 1 - Introduction

Before beginning a port scan, it is important to understand what a port represents.

A port is a logical communication endpoint used by applications and network services.

For example:

Port	Protocol / Service
20	FTP Data
21	FTP Control
22	SSH
23	Telnet
25	SMTP
53	DNS
80	HTTP
110	POP3
143	IMAP
443	HTTPS
445	SMB
3306	MySQL
3389	RDP

Port numbers are divided into three main ranges:

- Well-known ports: 0-1023
- Registered ports: 1024-49151
- Dynamic/private ports: 49152-65535

Port scanning helps determine which communication endpoints are accessible on a target machine.

Task 2 - TCP Connect Scan

TCP stands for Transmission Control Protocol.
It is a connection-oriented protocol that establishes communication between two devices through a three-way handshake.

The three-way handshake consists of:

i. SYN - The client requests a connection.
ii. SYN-ACK - The server acknowledges the request.
iii. ACK - The client completes the connection.

Nmap uses TCP Connect scanning to establish complete TCP connections with target ports.

Command:

```
nmap -sT TARGET_IP
```

To scan a specific port:

```
nmap -sT -p 80 TARGET_IP
```

To scan multiple ports:

```
nmap -sT -p 22,80,443 TARGET_IP
```

To scan a range of ports:

```
nmap -sT -p 1-1000 TARGET_IP
```

TCP Connect scans are useful when performing basic port enumeration, particularly when raw packet scanning is unavailable.

Task 3 - TCP SYN Scan

A TCP SYN scan is commonly referred to as a half-open scan.

It sends SYN packets to target ports without completing the entire TCP three-way handshake.

The scanning process:

i. Nmap sends a SYN packet.
ii. The target responds with SYN-ACK or RST depending on the port state.
iii. Nmap terminates the connection attempt with an RST when appropriate.

Command:

```
sudo nmap -sS TARGET_IP
```

To scan specific ports:

```
sudo nmap -sS -p 22,80,443 TARGET_IP
```

To scan all TCP ports:

```
sudo nmap -sS -p- TARGET_IP
```

SYN scanning is commonly used because it is efficient and does not complete a full TCP connection.

It typically requires elevated privileges on Linux.

Task 4 - UDP Port Scanning

UDP stands for User Datagram Protocol.

Unlike TCP, UDP does not establish a connection before sending data.

It is commonly used by services such as:

- DNS
- DHCP
- SNMP
- TFTP

- NTP

Nmap supports UDP scanning through the -sU option.

Basic UDP scan:

```
sudo nmap -sU TARGET_IP
```

Scan specific UDP ports:

```
sudo nmap -sU -p 53,161,162 TARGET_IP
```

Scan the most common UDP ports:

```
sudo nmap -sU --top-ports 100 TARGET_IP
```

UDP scans can be slower than TCP scans because many UDP services do not respond to unexpected requests.

Common UDP port states include:

State	Description
Open	A UDP response was received
Closed	The target returned an ICMP port-unreachable message
Filtered	Network filtering prevents a reliable determination
Open|Filtered	Nmap cannot distinguish between an open port and a filtered port

Task 5 - Port States

When Nmap performs a scan, it classifies ports based on the responses received.

Understanding these states is important when analysing scan results.

The six main Nmap port states are:

a. Open: An application is actively accepting connections or packets on the port.
b. Closed: The port is reachable, but no application is listening on it.
c. Filtered: Nmap cannot determine whether the port is open because packet filtering prevents a clear response.
d. Unfiltered: The port is accessible, but Nmap cannot determine whether it is open or closed.
e. Open | Filtered: Nmap cannot distinguish between an open port and a filtered port.
f. Closed | Filtered: Nmap cannot determine whether the port is closed or filtered. This state is used with specific scanning techniques.

These classifications help security professionals understand the accessibility of services and identify possible firewall restrictions.

Task 6 - Specifying Ports and Port Ranges

Nmap allows users to control which ports are scanned.

This is useful when performing targeted enumeration instead of scanning every available port.

Scan a single port

```
nmap -p 80 TARGET_IP
```

Scan multiple ports

```
nmap -p 22,80,443 TARGET_IP
```

Scan a range of ports

```
nmap -p 1-1000 TARGET_IP
```

Scan all TCP ports

```
nmap -p- TARGET_IP
```

Scan the most common ports

```
nmap --top-ports 100 TARGET_IP
```

Scan all ports using TCP SYN scanning

```
sudo nmap -sS -p- TARGET_IP
```

Exclude specific ports

```
nmap -p 1-1000 --exclude-ports 80,443 TARGET_IP
```

Selecting the correct port range can reduce scanning time and make enumeration more efficient.

Task 7 - Service and Version Detection

Discovering an open port is only the first step. We also need to identify which application or service is running on that port.

Nmap provides service and version detection through the -sV option.

Basic service detection:

```
nmap -sV TARGET_IP
```

Service detection on specific ports:

```
nmap -sV -p 22,80,443 TARGET_IP
```

Increase version detection intensity:

nmap -sV --version-intensity 7 TARGET_IP

Version detection can help identify:

- Service names.
- Application versions.
- Protocol information.
- Possible outdated software.
- Potentially vulnerable services.

For example, an open port might be associated with an SSH service, but version detection can provide additional information about the software implementation.

Task 8 - Nmap Scan Timing and Performance

Nmap provides timing templates that control scanning speed and timing behaviour.

There are six timing templates:

Template	Description
-T0	Paranoid
-T1	Sneaky
-T2	Polite
-T3	Normal
-T4	Aggressive
-T5	Insane

Example - Normal scan

nmap -T3 TARGET_IP

Example - Faster scan

nmap -T4 TARGET_IP

Example - Slower scan

nmap -T2 TARGET_IP

Timing templates affect scan performance and detection behaviour.
Faster scans may generate more network traffic and can be less reliable on unstable networks.

Task 9 - Detecting Services and Operating Systems

Nmap can combine multiple scanning techniques to collect detailed information about a target.
Operating system detection:
sudo nmap -O TARGET_IP
Service version detection:
nmap -sV TARGET_IP
Aggressive scanning:

sudo nmap -A TARGET_IP

The -A option enables several advanced features, including:

- OS detection.
- Version detection.
- Script scanning.
- Traceroute.

Aggressive scans are useful during authorized assessments when more detailed information about the target is required.

Task 10 - Saving Nmap Scan Results

Nmap provides several output formats that allow scan results to be saved for future analysis.

Normal output:

nmap -oN scan_results.txt TARGET_IP

XML output:

nmap -oX scan_results.xml TARGET_IP

Grepable output:

nmap -oG scan_results.gnmap TARGET_IP

Save in all major formats:

nmap -oA scan_results TARGET_IP

Saving scan results is useful when documenting penetration tests, comparing scan results, and preparing security assessment reports.

Task 11 - Practical Port Scanning

This section combines the scanning techniques covered in the room.

Use the target IP assigned by TryHackMe.

Exercise 1 - Basic Scan

Perform a basic Nmap scan.

nmap TARGET_IP

Observe the discovered ports and their states.

Exercise 2 - TCP Connect Scan

nmap -sT TARGET_IP

Compare the results with the basic scan.

Exercise 3 - SYN Scan

sudo nmap -sS TARGET_IP

Observe how Nmap identifies open and closed ports.

Exercise 4 - UDP Scan

sudo nmap -sU TARGET_IP

Examine how UDP port states differ from TCP port states.

Exercise 5 - Service Detection

nmap -sV TARGET_IP

Record the service names and version information.

Exercise 6 - Scan All TCP Ports

sudo nmap -sS -p- TARGET_IP

Compare the results with a default Nmap scan.

Exercise 7 - Save Scan Results

nmap -sV -oA basic_scan TARGET_IP

Inspect the generated output files.

5. Important Commands Cheat Sheet

Command	Purpose
nmap IP	Basic port scan
nmap -sT IP	TCP Connect scan
sudo nmap -sS IP	TCP SYN scan
sudo nmap -sU IP	UDP scan
nmap -p 80 IP	Scan a specific port
nmap -p 1-1000 IP	Scan a port range
nmap -p- IP	Scan all TCP ports
nmap --top-ports 100 IP	Scan common ports
nmap -sV IP	Service version detection
sudo nmap -O IP	Operating system detection
sudo nmap -A IP	Aggressive scan
nmap -T4 IP	Faster timing template
nmap -oN file.txt IP	Save normal output
nmap -oX file.xml IP	Save XML output
nmap -oA file IP	Save multiple output formats

6. Key Takeaways

After completing this room, you should understand:

- The purpose of network port scanning.
- How TCP Connect scans work.
- How TCP SYN scans work.
- How UDP scanning differs from TCP scanning.
- The meaning of different Nmap port states.
- How to scan individual ports and port ranges.
- How to identify running services and their versions.
- How Nmap timing templates affect scan performance.
- How to identify operating systems.
- How to save scan results for documentation.

7. Conclusion

The Basic Port Scans room introduces essential port scanning techniques using Nmap.

It explains how TCP and UDP scanning work, how to identify different port states, and how to gather additional information about services running on a target system.

These techniques are fundamental to penetration testing because identifying exposed ports and services helps security professionals understand the potential attack surface of a network.